AMIR MOUSAVI

BACKEND, AI SYSTEMS & MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

Over the past 10 years, I have developed a strong foundation in programming, engineering, mathematics, statistics, and problem-solving. Early in my career, I focused on mastering the principles behind effective software development. I believe strong software engineering extends far beyond writing code; it requires rigorous analytical thinking, well-considered system design, and a comprehensive understanding of the underlying problem, its constraints, and its broader context.

During the past five years, I have applied this foundation to developing software, data-driven tools, and artificial intelligence solutions for healthcare and medical research. Most recently, I built a self-hosted, multi-service document intelligence platform using Node.js, Next.js, React, Express, Python, PostgreSQL, Redis, OpenSearch, Docker, Nginx, and GitHub Actions. This project strengthened my practical experience in API design, asynchronous processing, authentication, database integration, search and retrieval, Linux deployment, continuous integration, and reproducible machine-learning workflows.

I continuously explore emerging technologies and challenge myself through complex projects that strengthen my ability to design, optimize, and deliver reliable, maintainable solutions. I am now prepared to contribute to large-scale, reliability-focused technology initiatives where I can combine software engineering, AI systems, innovation, and meaningful impact. For me, applying technology to solve complex systems problems and improve the tools engineers depend on is both professionally rewarding and personally fulfilling.

RESEARCH & TECHNICAL EXPERTISE

Programming & Scientific Computing	**Python, MATLAB, SQL, JavaScript**
Machine Learning & Research	**Machine learning, deep learning, computer vision, generative modelling, statistical learning, Bayesian methods, graph-based methods, optimization, cross-validation, model evaluation, feature engineering, experimental design, reproducible research**
Generative AI & Retrieval	**RAG, embeddings, vector search, keyword search, hybrid retrieval, scientific document intelligence, semantic chunking, grounded question answering, provenance-aware retrieval**
Data, Search & Systems	**PostgreSQL, MongoDB, Redis, OpenSearch, Elasticsearch, Linux, Docker, Git, GitHub Actions, CI/CD, REST APIs, asynchronous processing, deployment, debugging**
Backend & Web	**Node.js, Express, Django, Flask, Next.js, React**

PROFESSIONAL & RESEARCH EXPERIENCE

PhD Candidate / Research Software Developer - Université de Picardie Jules Verne (UPJV) — **2023 - Present**

Amiens, France | PhD contract through September 2026

- Develop reproducible analytical and machine-learning pipelines for stroke-patient and brain functional-connectivity data, spanning preprocessing, feature engineering, modelling, validation, visualization, and interpretation.
- Design and compare machine-learning, statistical, Bayesian, and graph-based approaches; apply optimization and multiple cross-validation strategies to improve robustness and generalizability.
- Develop reusable Python research tools and translate multidisciplinary research questions into computational formulations and experimental workflows.
- Collaborate across neuroscience, healthcare, statistics, engineering, and machine-learning domains; present methodological developments and research findings to multidisciplinary audiences.
- Presented PhD progress at two international conferences; one manuscript is currently under review as part of the thesis-defense authorization process.

Backend Developer - Maralhost — **May 2021 - Oct 2022**

Isfahan, Iran

- Developed and maintained Django backend functionality, server-side features, and data models for web applications in a hosting and server-services environment.
- Supported Linux-based deployment, debugging, database-backed applications, Git workflows, and web-security fundamentals.

Programmer & Electronics Designer - Mobarakeh Steel Company — **Oct 2017 - Oct 2019**

Mobarakeh, Isfahan, Iran

- Reverse-engineered and designed an electronic control board for regulating rolling-equipment speed; implemented control algorithms and evaluated system performance and reliability.

SELECTED SOFTWARE ENGINNERING PROJECT

DocuMind - Scientific Document Intelligence & RAG Platform | github.com/liamirpy/DocuMind

- Architected a self-hosted platform for organizing research PDFs, retrieving scientific information, and answering grounded questions with page-level citations.
- Developed workflows to extract text, figures, tables, formulas, references, metadata, and hierarchical document structure; generated semantic chunks and embeddings for downstream retrieval.
- Implemented vector, keyword, and hybrid retrieval with OpenSearch and integrated local model inference through Ollama for grounded scientific question answering.
- Designed provenance-aware, asynchronous, reproducible processing using Python services, PostgreSQL, Redis, OpenSearch, Docker, Nginx, and supporting web services.

PaperMind - Scientific PDF Processing Services | 2026 | github.com/liamirpy/PaperMind

- Developed Python services that convert scientific publications into structured, machine-readable document representations containing metadata, hierarchical sections, paragraphs, references, figures, tables, formulas, provenance, and quality information.
- Designed reusable API and command-line workflows for validation, indexing, retrieval, and RAG pipelines; integrated the services into DocuMind.

Baxter Chess-Playing Robot | 2026 | github.com/liamirpy/Baxter-chess-playing-Robot

- Developed a vision-based robotic system that detects human chess moves, reconstructs board state, plans legal responses, and commands a Baxter robot to move pieces physically.
- Integrated computer vision, chess-state reasoning, calibration, motion control, and hardware interaction; published source code and a demonstration video.

Medical Imaging & Deep Learning Research

- Developed deep-learning approaches for ischemic-stroke lesion segmentation from T1-weighted MRI and investigated lumbar-spine degeneration classification using computer-vision methods.
- Built a CT segmentation workflow for lung/infection lesions and a real-time face-mask detection system on Raspberry Pi for resource-constrained edge inference.
- Additional work includes generative eyes-to-face inpainting, Persian-language sentiment analysis, and a Deep Deterministic Policy Gradient implementation for secure wireless-communication research.

EDUCATION

PhD in Neuroscience - Candidate - Université de Picardie Jules Verne (UPJV) **2023 - Present**

Amiens, France | Contract through September 2026

- Research focus: machine learning and data-science methods for stroke and functional connectivity, statistical modelling, graph analysis, optimization, cross-validation, and reproducible scientific computing.

MSc in Electrical Engineering - Telecommunications - Shiraz University of Technology **2019 - 2023**

Shiraz, Iran

- Research thesis: Quantification of damaged lung tissue due to COVID-19 using computed tomography images.

BEng in Electrical Engineering - Telecommunications - University of Isfahan **2015 - 2019**

Isfahan, Iran

- Final project: Persian-language e-commerce sentiment analysis with positive/negative classification and negative-term extraction.

SELECTED PUBLICATIONS

- **Hassanpour, A.; Mousavi Mobarakeh, S. A.; Etefaghi Daryani, A.; Ramachandra, R.; Yang, B.** "Synthetic Face Generation Through Eyes-to-Face Inpainting." IEEE International Joint Conference on Biometrics (IJCB), 2023, 1-8. doi:10.1109/IJCB57857.2023.10449216.
- **Mousavi Mobarakeh, S. A.** "VRXU-net: A Deep Learning Approach for Brain Ischemic Stroke Lesion Detection and Segmentation in T1W MRI." arXiv 2605.21633, 2026.
- **Mousavi, A.** "RAX-NET: Residual Attention Xception Network for Brain Ischemic Stroke Segmentation in T1-Weighted MRI." medRxiv, 2025. doi:10.1101/2025.09.26.25336733.
- **Mousavi, A.** "Lumbar Spine Degenerative Classification Using YOLO v8 and DeepScoreNet." medRxiv, 2024. doi:10.1101/2024.12.06.24318595.
- **Mousavi Mobarakeh, S. A.; Kazemi, K.; Aarabi, A.; Danyali, H.** "Empowering Medical Imaging with Artificial Intelligence: A Review of Machine Learning Approaches for the Detection and Segmentation of COVID-19 Using Radiographic and Tomographic Images." arXiv 2401.07020, 2024.

Full publication list: Google Scholar

TEACHING, PRESENTATIONS & ADDITIONAL INFORMATION

Teaching: Teaching Assistant for approximately 60 hours of university-level Python and machine-learning instruction, laboratory support, and student guidance during 2025-2026.

Presentations: Presented PhD research progress and methodological findings at two international conferences to multidisciplinary scientific audiences.

Languages: Persian/Farsi (native); English (professional proficiency); French (A2 speaking, A1 writing, B1 listening comprehension).

Certifications: Advanced ARM Microcontroller (Metaco, 2018); AVR Microcontroller (Metaco, 2017).